```
clc
clear
close all
```

2

```
w=2e5;
count_s_6=0;
count_10=0;
s_first=0;
s_1=0;
for i=1:w
    j=1;
    k=1;
    t_s(1)=0;
    t_b(1)=0;
    count_s=0;
    count_b=0;

    while t_s(j)<5
        j=j+1;
        u=rand(1);
        t_s(j)=-log(u)./3+t_s(j-1);
        v=rand(1);
        if v<=.4
            count_s=count_s+1;
        end

    end
    while t_b(k)<5
        k=k+1;
        u=rand(1);
        t_b(k)=-log(u)./2+t_b(k-1);
        v=rand(1);
        if v<=.4
            count_b=count_b+1;
        end
    end
    if 10==count_b+count_s
        count_10=count_10+1;
        if 6==count_s
            count_s_6=count_s_6+1;
        end
    end
    if t_s(2)<t_b(2)
        s_first=s_first+1;
    end
    T_m(i)=t_b(2)+t_s(2);
    if t_s(2)<=1
        s_1=s_1+1;
    end
    c_b(i)=count_b;
```

```
    c_s(i)=count_s;
end
T_m=mean(T_m)
mean_s=mean(c_s)
mean_b=mean(c_b)
std_s=std(c_s)
std_b=std(c_b)
covariance=cov(c_s,c_b);
corr=covariance./(std_b.*std_s)
total=mean_s+mean_b
s_1_r=s_1/2e5
s_first_r=s_first/2e5
ratio_6_10=count_s_6/count_10
figure
hold on
histogram(c_s,'Normalization','pdf')
xline(mean_s,'-r','LineWidth',4)
xlabel('Number of arrivals from Stars')
ylabel('Probability density')
title('Number of photon arrivals from star')
hold off
figure
hold on
histogram(c_b,'Normalization','pdf')
xline(mean_b,'-r','LineWidth',4)
xlabel('Number of arrivals from Sky Background')
ylabel('Probability density')
title('Number of photon arrivals from sky')

hold off


%direct simulation
w=2e5;
lambda_s_d=3*.4;
lambda_b_d=2*.4;
for i=1:w

j=1;
k=1;
t_s_d(1)=0;
t_b_d(1)=0;

while t_s_d(j)<5
    j=j+1;
    u=rand(1);
    t_s_d(j)=-log(u)./lambda_s_d+t_s_d(j-1);


end
while t_b_d(k)<5
    k=k+1;
    u=rand(1);
    t_b_d(k)=-log(u)./lambda_b_d+t_b_d(k-1);
```

```
end
c_b_d(i)=k-1;
c_s_d(i)=j-1;
end
mean_s_d=mean(c_s_d)
mean_b_d=mean(c_b_d)
std_s_d=std(c_s_d)
std_b_d=std(c_b_d)
covariance=cov(c_s_d,c_b_d);
corr=covariance./(std_b_d.*std_s_d)


figure
hold on
histogram(c_s_d,'Normalization','pdf')
xline(mean_s_d,'-r','LineWidth',4)
xlabel('Number of arrivals from Stars')
ylabel('Probability density')
title('Number of photon arrivals from star (direct simulation)')
hold off

figure
hold on
histogram(c_b_d,'Normalization','pdf')
xline(mean_b_d,'-r','LineWidth',4)
xlabel('Number of arrivals from Sky Background')
ylabel('Probability density')
title('Number of photon arrivals from sky (direct simulation)')
hold off


T_m =

    0.8352


mean_s =

    6.3961


mean_b =

    4.4009


std_s =

    2.4991


std_b =
```

```
    2.0615


corr =

    1.2123    0.0040
    0.0040    0.8249


total =

   10.7970


s_1_r =

    0.9496


s_first_r =

    0.6006


ratio_6_10 =

    0.2575


mean_s_d =

    7.0003


mean_b_d =

    5.0052


std_s_d =

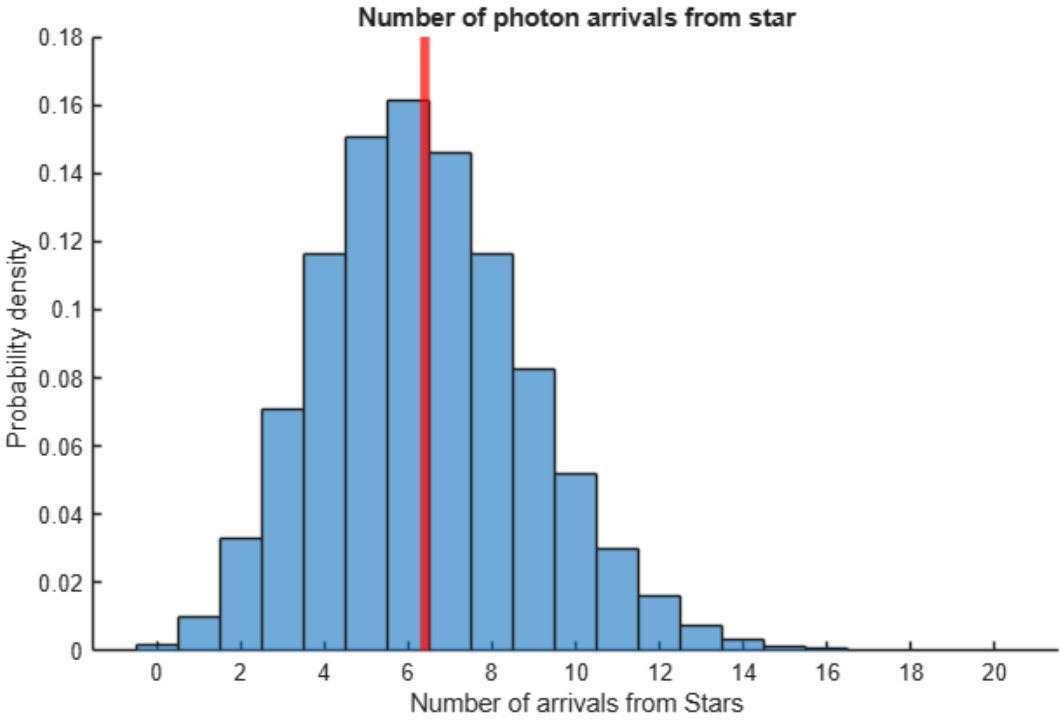
    2.4563

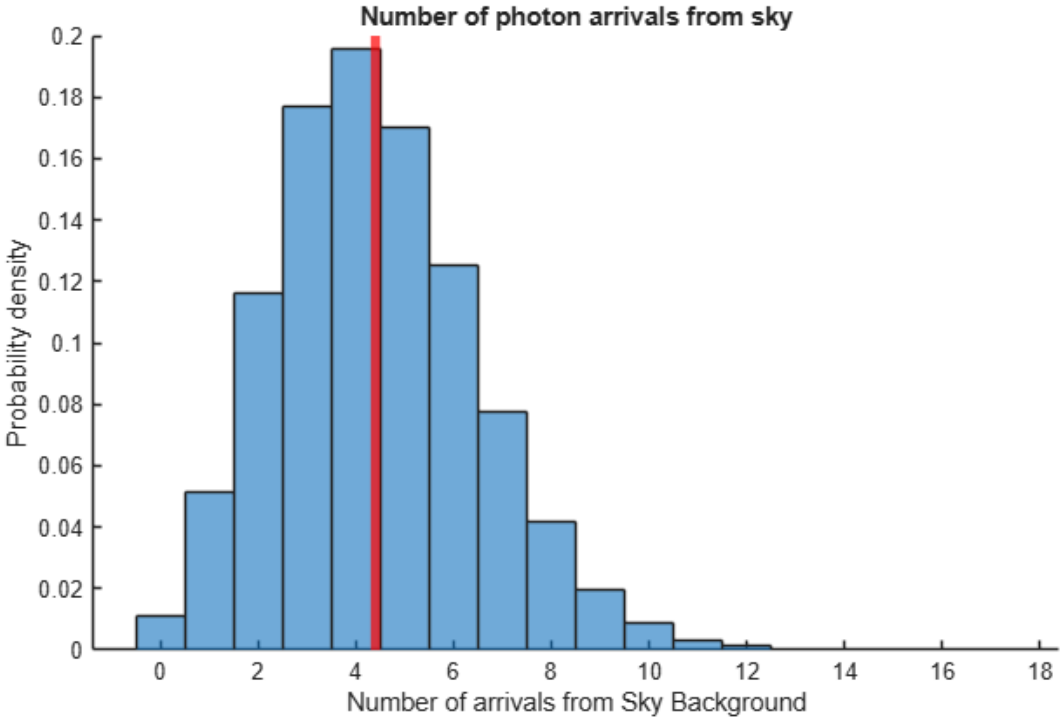

std_b_d =

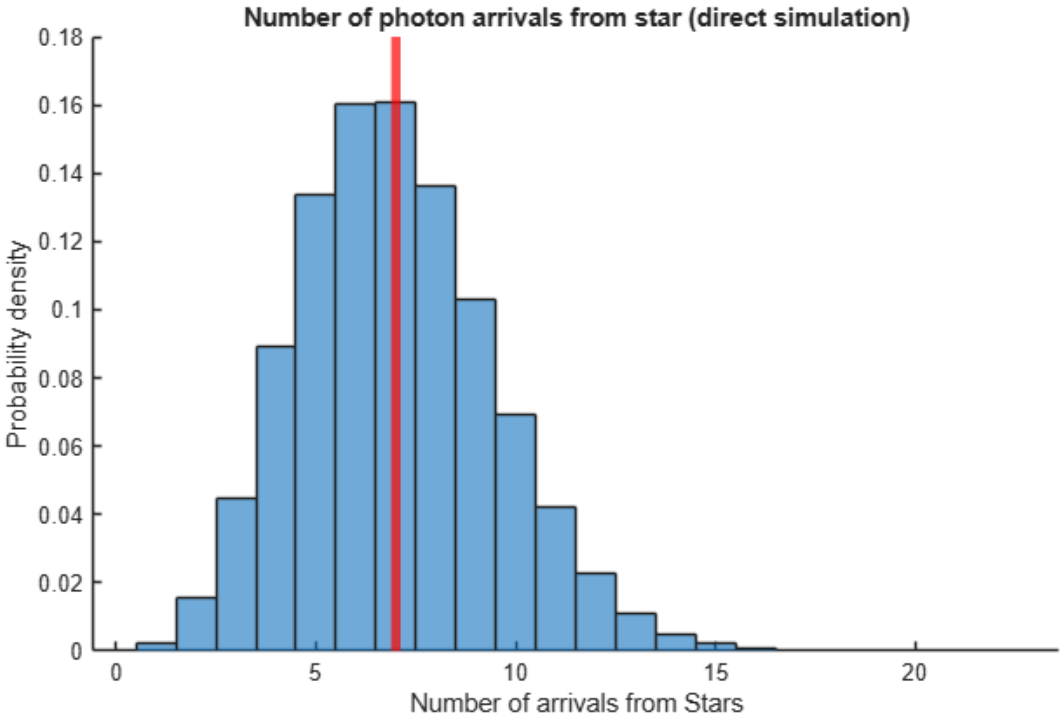
    2.0025

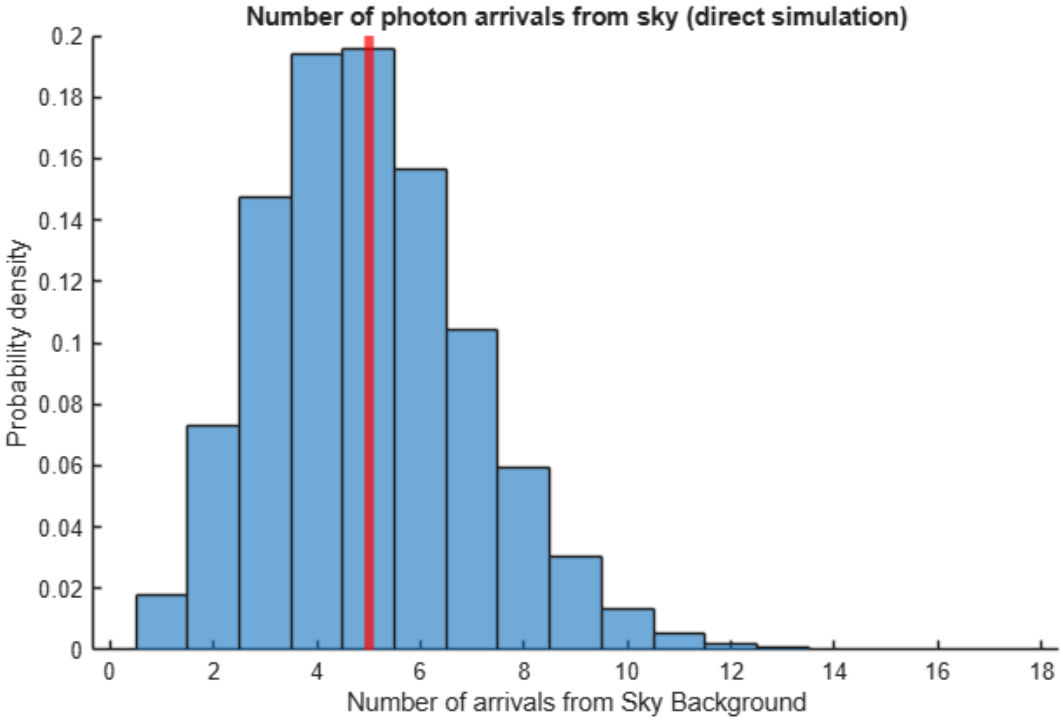

corr =

    1.2266   -0.0004
   -0.0004    0.8153
```


Number of photon arrivals from star
Probability density
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
2
4
6
8
10
12
14
16
18
20
Number of arrivals from Stars


Number of photon arrivals from sky
Probability density
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
2
4
6
8
10
12
14
16
18
Number of arrivals from Sky Background





3

```
%t=1:1./60:720;
j=1;
```

```
t(1)=0;
while t(j)<=720
    j=j+1;
    u=rand(1);
    t(j)=-log(u)./20+t(j-1);

end


count=0;
for i=1:length(t)
    u=rand(1);
    if u<rate(t(i))./20
        count=count+1;
        t_thin(count)=t(i);
    end
end
frac=count./length(t)
count_hour = zeros(1,720);
count_hour_analytical=zeros(1,720);
for i=1:720
   count_hour(i)=sum(t_thin>=(i-1) & t_thin<i);
   count_hour_analytical(i)=def_int(i-1,i);
   n=0:29;
   for k=1:length(n)
       arrivals_34(k)=sum(t_thin>=3+24.*n(k) & t_thin<4+24.*n(k));
   end
end
mean_arrivals_34 = mean(arrivals_34);
std_arrivals_34 = std(arrivals_34);
fprintf('Mean arrivals from 3:00 to 4:00: %.3f\n', mean_arrivals_34);
mean_poisson=4.091;
var_poisson=sqrt(4.091)

for l=1:720
sum_rate=sum(log(rate(1:l)));
maximized_log(l)=sum_rate-def_int(0,720);
end
figure
hold on
plot(1:720,rate(1:720))
plot(1:720,maximized_log)


figure
hold on
plot(1:720,count_hour,'-*')
plot(1:720,count_hour_analytical,'-' )
plot(1:720,rate(1:720),'--')
title('Thinned Counts Versus Analytical Counts')
legend('Thinned Counts','Rate Function','Integral of Rate Function')
xlabel('Hours')
ylabel('Number of Arrivals')
hold off
```

```
figure
plot(1:720, (count_hour-rate(1:720))./20)
function lambda_t=rate(t)
    lambda_t=12+8.*cos(pi.*(t-16)./12);
end
function N=int(t)
    N=12.*t+96.*sin(pi.*(t-16)./12)./pi;
end
function N=def_int(s,t)
    N=int(t)-int(s);
end


sum=0;
for i=0:10
    sum=sum+(7.1438^(i))*exp(-7.1438)/(factorial(i));
end
1-sum


sum=0;
for i=0:9
    sum=sum+factorial(i+19)*((12/(12+5))^19)*((5/(5+12))^i)/
(factorial(19)*factorial(i))
    %sum=sum+(10^(i))*exp(-10)/(factorial(i));
end
1-sum


frac =

    0.5986

Mean arrivals from 3:00 to 4:00: 4.467

var_poisson =

    2.0226


ans =

    0.1090


sum =

    0.0013


sum =

    0.0092
```

```
sum =

    0.0335

sum =

    0.0858

sum =

    0.1744

sum =

    0.2994

sum =

    0.4526

sum =

    0.6200

sum =

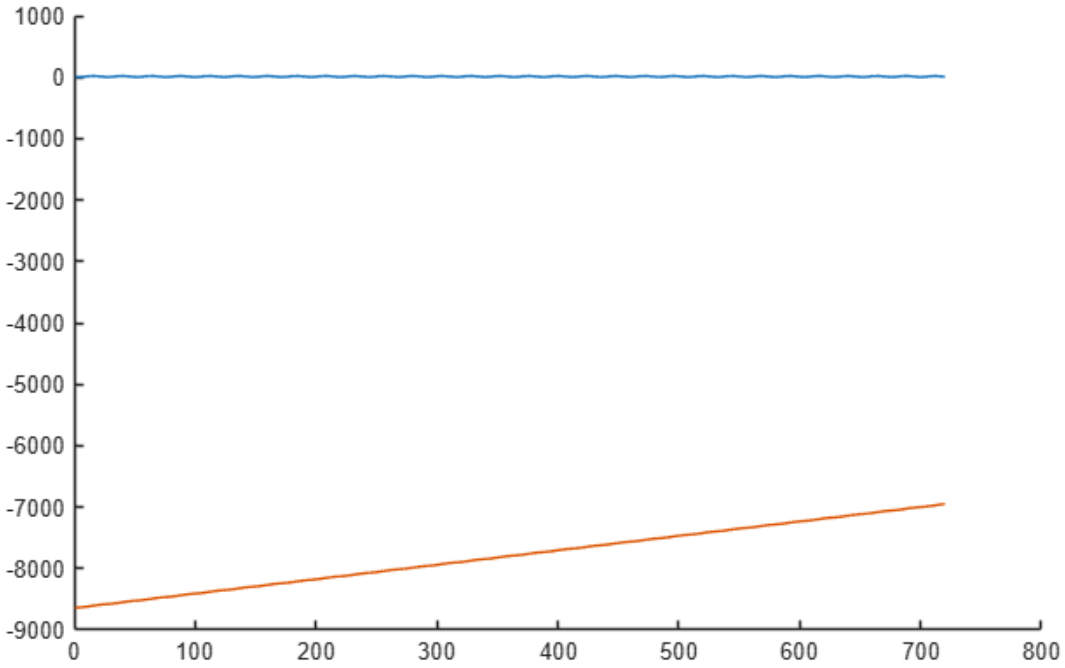
    0.7861

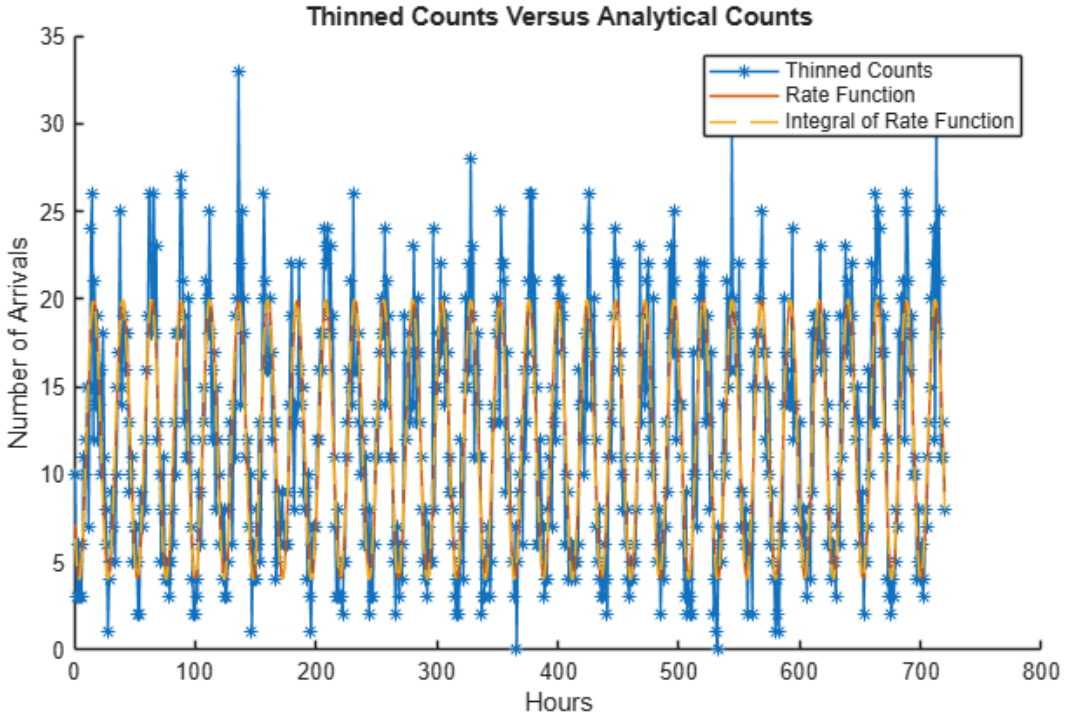
sum =

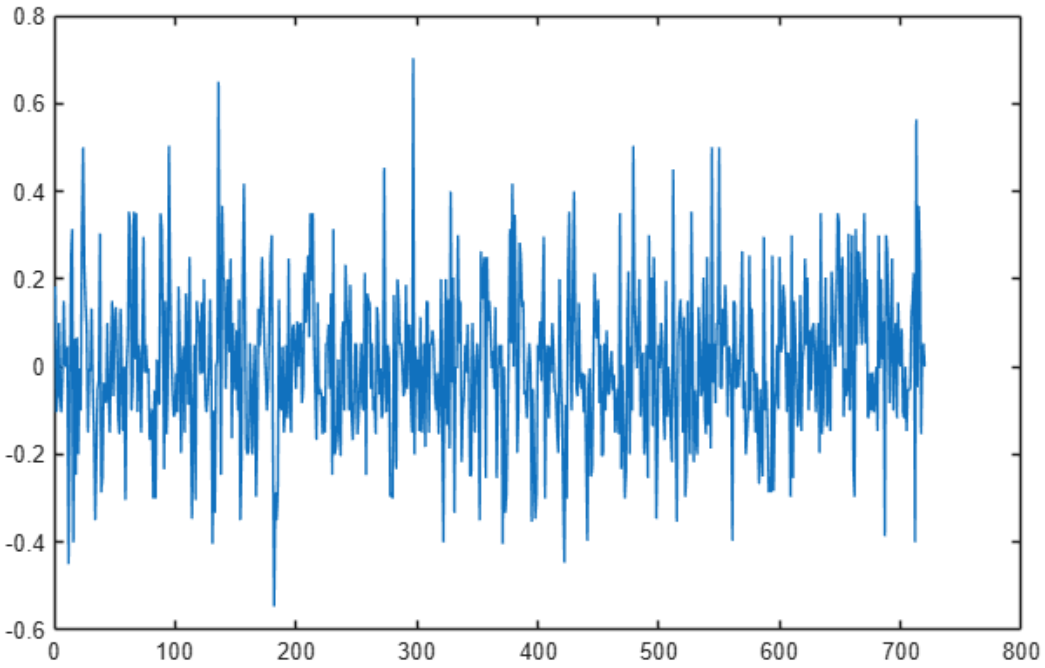
    0.9381

ans =

    0.0619
```


1000
0
-1000
-2000
-3000
-4000
-5000
-6000
-7000
-8000
-9000
0
100
200
300
400
500
600
700
800


Thinned Counts Versus Analytical Counts
Thinned Counts
Rate Function
Integral of Rate Function
Number of Arrivals
Hours
35
30
25
20
15
10
5
0
0
100
200
300
400
500
600
700
800



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